ACKNOWLEDGEMENT

I would like to express my sincere gratitude to all those who supported me in the completion of this seminar report.

First and foremost, I am deeply grateful to my **Mini Project Guide, Prof. Rushali Katkar**, for their invaluable guidance, constant encouragement, and thoughtful feedback throughout this work. Their expertise in Machine Learning and Data Science greatly shaped the direction and depth of this report.

I would like to acknowledge the open-source community and contributors to libraries such as Scikit-learn, XGBoost, Pandas, NumPy, Seaborn, Matplotlib, and Imbalanced-learn, whose tools formed the backbone of this project's implementation.

Finally, I am sincerely thankful to my family and friends for their unwavering moral support and encouragement throughout this journey.

Ashish Bajpai
7314

ABSTRACT

In the modern insurance industry, cross-selling represents one of the most efficient strategies for maximizing customer lifetime value and reducing customer acquisition costs. Health insurance providers, who already possess rich demographic and behavioural data about their policyholders, are uniquely positioned to leverage this data for predicting interest in complementary products such as vehicle insurance. This seminar report presents a comprehensive investigation into the application of supervised machine learning algorithms to the problem of Health Insurance Cross-Sell Prediction.

The study utilizes a large-scale real-world dataset comprising 381,109 customer records with 12 features encompassing demographic attributes, vehicle characteristics, and prior insurance history. A key challenge identified in the dataset is a severe class imbalance, where only approximately 12.25% of customers express interest in vehicle insurance. This imbalance, if unaddressed, causes predictive models to become biased toward the majority class. To counter this, the study employs the **Random Over-Sampling (ROS)** technique from the Imbalanced-learn library, which effectively equalizes class representation before model training.

Three distinct classification algorithms are implemented and rigorously evaluated: **Logistic Regression**, **Random Forest Classifier**, and the **XGBoost Classifier**. Comprehensive Exploratory Data Analysis (EDA) is conducted prior to modelling to uncover relationships between predictor variables and the target response. Fea-

ture selection using the Extra Trees Classifier identifies Age, Annual Premium, Vintage, Previously Insured, and Vehicle Damage History as the most influential predictors.

Model performance is assessed using a multi-metric framework including Accuracy, Precision, Recall, F1-Score, and the ROC-AUC score. Experimental results demonstrate that the **Random Forest Classifier** achieves superior performance with an accuracy of 93.91% and an outstanding ROC-AUC score of 0.9979, significantly outperforming Logistic Regression (AUC: 0.8219) and XGBoost (AUC: 0.8684). The findings of this study provide a robust and deployable framework for insurance companies to prioritize their marketing communication and optimize revenue generation through intelligent, data-driven targeting.

Contents

Acknowledgement **3**

Abstract **4**

1. Introduction **9**

1.1. General 9

1.2. Problem Definition 9

2. Literature Review **11**

2.1. General 11

2.2. Machine Learning in Insurance Analytics 11

2.3. Cross-Sell Prediction in Financial Services 12

2.4. Handling Class Imbalance 12

2.5. Feature Selection Techniques 12

3. Dataset Description and Exploratory Data Analysis **13**

3.1. Dataset Description 13

3.1.1. Data Quality Assessment 14

3.1.2. Descriptive Statistics 15

3.2. Exploratory Data Analysis 16

3.2.1. Target Variable Distribution 16

3.2.2. Gender vs. Response 16

3.2.3. Age vs. Response 16

3.2.4. Driving Licence vs. Response 17

3.2.5. Previously Insured vs. Response 17

3.2.6. Vehicle Age vs. Response 18

3.2.7. Annual Premium Distribution 18

4. Methodology and Experimentation **19**

4.1. Overview of Methodology 19
4.2. Data Preprocessing 19
4.2.1. Label Encoding of Categorical Variables 19
4.2.2. Duplicate Row Detection 20
4.3. Feature Selection 20
4.4. Handling Class Imbalance: Random Over-Sampling 21
4.5. Train-Test Split and Feature Scaling 21
4.6. Model Selection Rationale 22
4.6.1. Model 1: Logistic Regression 23
4.6.2. Model 2: Random Forest Classifier 23
4.6.3. Model 3: XGBoost Classifier 23

5. Results and Discussions 24

5.1. Evaluation Metrics Framework 24
5.2. Model 1: Logistic Regression Results 25
5.2.1. Performance Metrics 25
5.2.2. Classification Report 25
5.2.3. Analysis 26
5.3. Model 2: Random Forest Classifier Results 26
1. Performance Metrics 27
2. Classification Report 27
3. Analysis 27
4. Model 3: XGBoost Classifier Results 28
4.1. Performance Metrics 28
4.2. Classification Report 28
4.3. Analysis 28
5. Confusion Matrix Analysis 29

1. Conclusion and Scope for Future Work **30**

1.1. Conclusion 30

1.2. Scope for Future Work 31

References **33**

C H A P T E R 1

INTRODUCTION

1. GENERAL

The insurance sector is one of the most data-rich industries in the world. Every interaction between a customer and an insurance company — policy purchase, claim filing, premium payment, or a customer service call — generates data that, when analysed intelligently, can reveal patterns of behaviour, preference, and risk. With the proliferation of digital platforms and the advent of powerful machine learning algorithms, insurance companies are increasingly turning to data science to solve fundamental business challenges: customer retention, fraud detection, risk pricing, and most relevantly, cross-selling.

Cross-selling refers to the practice of offering additional or complementary products to existing customers. In the insurance context, a health insurance company may seek to sell vehicle insurance to its existing policyholders. This is a more cost-effective strategy than acquiring entirely new customers, as the company already has established a trust relationship and possesses detailed information about the customer.

2. PROBLEM DEFINITION

An insurance company has provided health insurance to a large pool of customers. Now, the company wishes to introduce vehicle insurance and wants to identify which of its existing health insurance customers are likely to be interested in this new offering. Formally, the problem is defined as a **binary classification task**:

- **Output = 1**: The customer is interested in vehicle insurance.
- **Output = 0**: The customer is not interested in vehicle insurance.

The input to the model consists of customer attributes including age, gender, driving licence status, vehicle age, vehicle damage history, annual premium paid for health insurance, number of days since policy commencement (vintage), and region code. The model must learn the relationship between these features and the target response variable with sufficient accuracy to be practically deployable in the company's marketing pipeline.

C H A P T E R 2

LITERATURE REVIEW

1. GENERAL

The intersection of machine learning and the insurance industry has been an active area of research in recent years. Researchers have explored a variety of problems including insurance fraud detection, claims prediction, customer churn, risk classification, and product recommendation. This chapter surveys the most relevant prior works that inform the methodology adopted in this seminar.

2. MACHINE LEARNING IN INSURANCE ANALYTICS

Traditional actuarial methods in the insurance industry relied heavily on statistical regression models and expert judgment. While these methods are interpretable and well-understood, they often fail to capture complex non-linear interactions between variables that are typical in large customer datasets.

Charpentier et al. (2018) demonstrated that machine learning models, particularly ensemble methods such as Random Forests and Gradient Boosting Machines, consistently outperform linear models on insurance pricing and risk prediction tasks. They attributed this to the ability of ensemble methods to model complex interactions without requiring manual feature engineering.

3. CROSS-SELL PREDICTION IN FINANCIAL SERVICES

Cross-selling recommendation systems in banking and insurance have been studied from multiple perspectives. **Hartfeil (1996)** proposed early rule-based systems for bank product cross-selling based on customer transaction data. Modern approaches have since moved to deep learning and gradient boosting frameworks.

Li et al. (2020) proposed a hybrid deep learning architecture combining recurrent neural networks (RNN) for sequence modelling with attention mechanisms for predicting financial product uptake. While their model achieved high accuracy, its interpretability was limited, which is a significant concern in regulated industries like insurance.

4. HANDLING CLASS IMBALANCE

Class imbalance is a well-documented challenge in binary classification problems, particularly in domains like fraud detection, medical diagnosis, and insurance response prediction where the event of interest is rare.

Chawla et al. (2002) introduced the Synthetic Minority Over-sampling Technique (SMOTE), which generates synthetic samples for the minority class by interpolating between existing minority samples. SMOTE has become one of the most widely used techniques for addressing class imbalance.

5. FEATURE SELECTION TECHNIQUES

Selecting the most informative features from a high-dimensional dataset is critical for both model performance and interpretability. **Geurts et al. (2006)** introduced **Extra Trees (Extremely Randomised Trees)**, an ensemble method that computes feature importance scores by averaging the impurity decrease across all trees in the forest.

CHAPTER 3

DATASET DESCRIPTION AND EXPLORATORY DATA ANALYSIS

1. DATASET DESCRIPTION

The dataset used in this study is sourced from an insurance company's historical records. It contains **381,109 rows** and **12 columns**, representing individual customers and their associated attributes. Table 3.1 provides a complete description of all features.

Table 3.1: Dataset Feature Description

Feature	**Data Type**	**Description**
id	Integer	Unique identifier for each customer
Gender	Categori ca l	Gender of the customer (Male/Female)
Age	Integer	Age of the customer in years
Driving_License	Binary	1 if the customer holds a valid driving li- cence, 0 otherwise
Region_Code	Float	Unique region code where the customer resides
Previously_Insured	Binary	1 if the customer already has vehicle insurance, 0 otherwise

Vehicle_Damage	Categori ca l	Whether the customer's vehicle was dam- aged in the past (Yes/ No)
Annual_Premium	Float	Annual premium (in INR) paid for the health insurance policy
Policy_Sales_Chan ne l	Float	Anonymised code for the channel through which the policy was sold
Vintage	Integer	Number of days the customer has been associated with the company
Response	Binary	**Target variable**: 1 if interested in vehicle insurance, 0 otherwise

1.1. Data Quality Assessment

A thorough data quality check was performed, yielding the following observations:

- **Missing Values:** No missing values were found in any of the 12 columns, confirming the dataset's completeness and eliminating the need for imputation strategies.

- **Duplicate Records:** A duplicate check revealed no redundant rows in the dataset, ensuring all 381,109 records represent unique customers.

- **Data Types:** After initial inspection, three columns were identified as requiring encoding: `Gender` (object), `V1e4hicle_Age` (object), and `Vehicle_Damage` (object). All remaining columns were already in numeric format.

1.2. Descriptive Statistics

Table 3.2: Descriptive Statistics of Numerical Features

Feature	Mean	Std Dev	Min	Median	Max
Age	38.82	15.51	20	36	85
Annual_Premi u m	30,564.39	17,213.15	2,630	31,669	540,165
Vintage	154.35	83.67	10	154	299
Region_Code	26.39	13.23	0	28	52

2. EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis (EDA) was conducted to understand the distribution of individual features, their relationship with the target variable, and inter-feature correlations. This analysis is critical for making informed preprocessing decisions.

2.1. Target Variable Distribution

The target variable `Response` is distributed as follows:

- **Class 0 (Not Interested):** 334,399 customers (87.75%)
- **Class 1 (Interested):** 46,710 customers (12.25%)

This represents an approximate **7.16:1 class imbalance** ratio, where the majority class significantly dominates. A count plot reveals this stark disparity visually, confirming that class balancing is essential before model training. Without intervention, a naive classifier that always predicts Class 0 would achieve 87.75% accuracy — a misleadingly high figure that masks total failure on the minority class.

2.2. Gender vs. Response

The gender distribution in the dataset is nearly equal, with males slightly outnumbering females. Analysis of the response variable by gender reveals that:

- Males are slightly more likely to express interest in vehicle insurance than females.
- The gender imbalance is marginal and does not constitute a dominant predictor on its own.

2.3. Age vs. Response

Age is one of the most informative features in the dataset. Key observations include:

- **Younger customers (below age 25)** show very low interest in vehicle insurance. This can be attributed to lower vehicle ownership rates, fewer assets, and generally lower risk awareness among younger demographics.
- **Middle-aged customers (ages 30–60)** show the highest interest rates. This demographic typically owns vehicles of significant value and has more experience with the financial consequences of vehicle damage.
- **Older customers (above 60)** show a declining interest, possibly due to reduced vehicle usage or preference for existing insurance arrangements.
- A boxplot of the Age variable confirms **no outliers**, making age a clean and reliable predictor.

2.4. Driving Licence vs. Response

Almost all customers who expressed interest in vehicle insurance (Response = 1) hold a valid driving licence. This is logically expected — a person who drives a vehicle is far more likely to seek vehicle insurance than one who does not drive. Customers without a driving licence show negligible interest.

2.5. Previously Insured vs. Response

The `Previously_Insured` feature demonstrates a very clear and intuitive pattern:

- Customers who are **not currently insured** (Previously_Insured = 0) are significantly more likely to be interested in purchasing vehicle insurance.

- Customers who **already have vehicle insurance** (Previously_Insured = 1)

- show virtually no interest, which is entirely logical — they have no reason to purchase a product they already own.

This feature is likely to be one of the strongest predictors in the model.

2.6. Vehicle Age vs. Response

The age of the customer's vehicle is categorised into three groups. The response distribution across these groups reveals:

- Vehicles aged **1–2 years** generate the most interest in insurance. Owners of relatively new but used vehicles may perceive significant residual value worth protecting.
- Vehicles aged **less than 1 year** generate the least interest — possibly because new vehicles often come with manufacturer warranties or dealer insurance arrangements.
- Vehicles **older than 2 years** generate moderate interest.

2.7. Annual Premium Distribution

The `Annual_Premium` variable exhibits a distinctly **right-skewed distribution**, as evidenced by its histogram with Kernel Density Estimate (KDE). The mean (INR 30,564) is significantly higher than the median (INR 31,669), with a long tail extending toward very high premium values.

A boxplot of `Annual_Premium` reveals the presence of **numerous outliers** at the upper end of the distribution. While these outliers were not removed (they likely represent genuine high-premium customers), the application of **Standard-Scaler** during preprocessing effectively mitigates their distorting effect on model training.

C H A P T E R 4

METHODOLOGY AND EXPERIMENTATION

1. OVERVIEW OF METHODOLOGY

The experimental methodology followed in this study is structured as a standard machine learning pipeline:

1. Library imports and environment setup
2. Dataset loading and initial inspection
3. Exploratory Data Analysis (EDA)
4. Data preprocessing (encoding, duplicate check, outlier assessment)
5. Feature selection using Extra Trees Classifier
6. Class balancing using Random Over-Sampling
7. Train-test split and feature scaling
8. Model training (Logistic Regression, Random Forest, XGBoost)
9. Model evaluation (Accuracy, Precision, Recall, F1-Score, ROC-AUC, Confusion Matrix)

2. DATAPREPROCESSING

1. Label Encoding of Categorical Variables

Three categorical features in the dataset — `Gender`, `Vehicle_Age`, and `Vehicle_Damage` — were converted to numerical representations using **Label Encoding**. This is a necessary step since most machine learning algorithms operate on numerical inputs.

```
labelEncoder = LabelEncoder()
data['Gender'] = labelEncoder.fit_transform(data['Gender'])
data['Vehicle_Age'] = labelEncoder.fit_transform(data['Vehicle_Age
    '])
data['Vehicle_Damage'] = labelEncoder.fit_transform(data['
    Vehicle_Damage'])
```

After encoding:

- `Gender`: Female → 0, Male → 1
- `Vehicle_Damage`: No → 0, Yes → 1
- `Vehicle_Age`: 1 Year → 0, 1–2 Year → 1, 2 Years → 2

2. Duplicate Row Detection

A duplicate row check was performed to ensure data integrity:

```
duplicate = data[data.duplicated()]
print(duplicate)  # Output: Empty DataFrame -- no duplicates found
```

Listing 4.3: Duplicate Row Check

The output confirmed that the dataset contains **no duplicate entries**.

3. FEATURE SELECTION

Feature selection was performed using the **Extra Trees Classifier**, which computes feature importance scores based on the mean decrease in impurity across all decision trees in the ensemble.

4. HANDLING CLASS IMBALANCE: RANDOM OVER-SAMPLING

Given the 7.16:1 class imbalance identified in the dataset, **Random Over-Sampling (ROS)** was applied to the training feature set and target vector prior to the train-test split. ROS randomly duplicates samples from the minority class (Response = 1) until both classes are equally represented.

```
x = data.drop(['Response', 'Driving_License', 'Gender', 'id'],
    axis=1)
y = data['Response']

randomsample = RandomOverSampler()
x_new, y_new = randomsample.fit_resample(x, y)

print('Original dataset shape:', Counter(y))
# Output: Counter({0: 334399, 1: 46710})

print('Resampled dataset shape:', Counter(y_new))
# Output: Counter({0: 334399, 1: 334399})
```

After resampling, both classes contain **334,399 samples each**, resulting in a balanced dataset of **668,798 total records** for model training.

5. TRAIN-TEST SPLIT AND FEATURE SCALING

The balanced dataset was divided into training (70%) and testing (30%) subsets using a stratified random split with `random_state=0` for reproducibility.

```
xtrain, xtest, ytrain, ytest =
    train_test_split( x_new, y_new, test_size=0.30,
    random_state=0
)
# xtrain shape: (468158, 9)
# xtest shape:  (200640, 9)

scaler = StandardScaler()
xtrain = scaler.fit_transform(xtrain)
xtest  = scaler.transform(xtest)
```

Standard Scaling (z-score normalisation) was applied to ensure all features are on a comparable scale, which is particularly important for Logistic Regression, which is sensitive to the magnitude of input features. The scaler was fitted exclusively on the training data and then applied to the test data to prevent data leakage.

6. MODEL SELECTION RATIONALE

The problem is identified as a **binary classification** task on a large dataset (records).The following considerations guided model selection:

- **Support Vector Machine (SVM)** was excluded due to its to maintaining complexity, which is computationally prohibitive for datasets of this scale.
- **Logistic Regression** was selected as a simple, interpretable baseline model that provides probabilistic outputs.

- **Random Forest Classifier** was selected for its ability to handle non-linear relationships, resistance to overfitting through bagging, and built-in feature importance.
- **XGBoost Classifier** was selected for its efficiency, strong regularisation capabilities, and proven performance on tabular binary classification tasks.

1. Model 1: Logistic Regression

Logistic Regression is a linear probabilistic classifier that models the log-odds of the positive class as a linear combination of the input features:

2. Model 2: Random Forest Classifier

The Random Forest algorithm constructs a multitude of decision trees at training time and outputs the class that is the mode of the classes predicted by the individual trees. It uses **bagging** (Bootstrap Aggregating) to train each tree on a random subset of the data, and at each node, considers only a random subset of features. This dual randomisation reduces variance significantly compared to a single decision tree.

3. Model 3: XGBoost Classifier

XGBoost (Extreme Gradient Boosting) is a scalable, distributed gradient boosting algorithm. Unlike Random Forest which builds trees in parallel, XGBoost builds trees **sequentially**, where each new tree corrects the errors of the previous ones by

fitting the residuals. It uses second-order Taylor expansion of the loss function to compute gradients efficiently and includes L1 and L2 regularisation terms to prevent overfitting.

C H A P T E R 5

RESULTS AND DISCUSSIONS

1. EVALUATION METRICS FRAMEWORK

All three models were evaluated using the following metrics, chosen to provide a comprehensive view of performance on an originally imbalanced binary classification problem:

Accuracy: Proportion of total correct predictions. Can be misleading with imbalanced classes

- **Recall (Sensitivity):** Out of all actual positive cases, what fraction was correctly identified.
- **F1-Score:** Harmonic mean of Precision and Recall, balancing both concerns.
- **ROC-AUC Score:** Area under the Receiver Operating Characteristic curve. Measures the model's ability to discriminate between the two classes across all classification thresholds. A score of 1.0 is perfect; 0.5 is rand

2. MODEL 1: LOGISTIC REGRESSION RESULTS

1. Performance Metrics

The Logistic Regression model achieved the following results on the test set:

Table 5.1: Logistic Regression – Performance Metrics

Metric	Value
Accuracy	0.7836 (78.36%)
Precision	0.7044
Recall	0.9772
F1-Score	0.8187
ROC-AUC Score	0.8219

2. Classification Report

Table 5.2: Logistic Regression – Classification Report

Class	Precision	Recall	F1-Score	Support
0 (Not Interested)	0.96	0.59	0.73	100,372
1 (Interested)	0.70	0.98	0.82	100,268
Weighted Avg	**0.83**	**0.78**	**0.77**	200,640

3. Analysis

Logistic Regression achieves a very high Recall (0.9772) for the positive class, meaning it successfully identifies approximately 97.7% of customers who are genuinely interested. However, its Precision is considerably lower (0.7044), indicating a significant number of false positives — customers predicted to be interested who are actually not. This trade-off is reflected in the moderate F1-Score and AUC.

The low precision can be attributed to Logistic Regression's assumption of linear decision boundaries, which may be insufficient to capture the complex non-linear patterns in the data.

3. MODEL 2: RANDOM FOREST CLASSIFIER RESULTS

1. Performance Metrics

Table 5.3: Random Forest Classifier – Performance Metrics

Metric	Value
Accuracy	0.9391 (93.91%)
Precision	0.8936
Recall	0.9971
F1-Score	0.9425
ROC-AUC Score	0.9979

2. Classification Report

Table 5.4: Random Forest – Classification Report

Class	Precision	Recall	F1-Score	Support
0 (Not Interested)	1.00	0.88	0.94	100,372
1 (Interested)	0.89	1.00	0.94	100,268
Weighted Avg	**0.94**	**0.94**	**0.94**	200,640

3. Analysis

The Random Forest Classifier delivers outstanding performance across all metrics. Its ROC-AUC score of **0.9979** is near-perfect, indicating exceptional discriminatory power between interested and non-interested customers. The combination of extremely high Recall (0.9971) and substantially improved Precision (0.8936) relative to Logistic Regression demonstrates that Random Forest effectively learns the complex non-linear boundaries in the data. The ROC curve for this model closely hugs the top-left corner, confirming superior performance.

5.3 MODEL 3: XGBOOST CLASSIFIER RESULTS

1. Performance Metrics

Table 5.5: XGBoost Classifier – Performance Metrics

Metric	Value
Accuracy	0.8118 (81.18%)
Precision	0.7499
Recall	0.9356
F1-Score	0.8325
ROC-AUC Score	0.8684

2. Classification Report

Table 5.6: XGBoost – Classification Report

Class	Precision	Recall	F1-Score	Support
0 (Not Interested)	0.94	0.69	0.79	100,372
1 (Interested)	0.75	0.94	0.83	100,268
Weighted Avg	**0.84**	**0.81**	**0.81**	200,640

3. Analysis

XGBoost performs better than Logistic Regression but falls short of Random Forest in this particular experimental setting. Its AUC of 0.8684 and accuracy of 81.18% suggest it captures more non-linearity than Logistic Regression but may benefit from hyperparameter tuning (e.g., adjusting `n_estimators`, `max_depth`, `learning_rate`, `subsample`) to reach its full potential.

2. CONFUSION MATRIX ANALYSIS

A confusion matrix provides a breakdown of correct and incorrect predictions into four categories: True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN).

Model	TP	TN	FP	FN
Logistic Regression	98,030	58,939	41,433	2,238
Random Forest	99,968	88,330	12,042	300
XGBoost	93,830	68,880	31,492	6,438

The Random Forest model produces the fewest False Negatives (300) — customers who are actually interested but are incorrectly classified as not interested. This is critical in a business context: a low FN rate means fewer genuinely interested customers are missed, maximising the reach of the marketing campaign.

CHAPTER 6

CONCLUSION AND SCOPE FOR FUTURE WORK

1. CONCLUSION

This seminar report has presented a comprehensive study on the application of supervised machine learning techniques to the problem of Health Insurance Cross-Sell Prediction. The goal was to build a predictive model that could identify, from among existing health insurance customers, those most likely to be interested in purchasing vehicle insurance.

The study successfully achieved its objectives through a rigorous and systematic approach:

1. A large-scale real-world dataset of **381,109 customer records** was acquired, loaded, and inspected. The dataset was found to be complete with no missing values or duplicate records.

2. **Exploratory Data Analysis** revealed rich patterns: middle-aged customers (30–60 years), those without existing vehicle insurance, those with prior vehicle damage, and those owning vehicles aged 1–2 years are the most likely targets for cross-selling.

3. The severe **class imbalance** (87.75% vs. 12.25%) was effectively addressed using **Random Over-Sampling**, creating a balanced training set that enabled unbiased model learning.

4. **Random Forest Classifier** emerged as the clear winner, demonstrating near-perfect discriminatory performance with an ROC-AUC of 0.9979, making it highly suitable for deployment in the insurance company's marketing pipeline.

From the analysis, the following actionable conclusions can be drawn for the insurance business:

- **Primary Target Segment:** Customers aged 30–60, without existing vehicle insurance, with a history of vehicle damage, and owning a vehicle aged 1–2 years.
- **Secondary Insight:** Annual Premium paid for health insurance is a moderate predictor, suggesting that customers who invest more in health protection may also be inclined to protect their vehicles.

2. SCOPE FOR FUTURE WORK

While the current study demonstrates strong results, several avenues exist for further improvement and extension:

1. **Hyperparameter Tuning:** Systematic tuning of model hyperparameters using **Grid Search Cross-Validation** or **Bayesian Optimisation** (e.g., using `Optuna` or `Hyperopt`) could further improve XGBoost and potentially achieve performance comparable to Random Forest.
2. **Deep Learning Approaches:** Architectures such as **Feedforward Neural Networks (FNN)** and **TabNet** (a transformer-based model designed for tabular data) could be explored for this task, potentially capturing higher-order feature interactions.

3. **Advanced Sampling Techniques:** Alternatives to simple Random Over-Sampling, such as **SMOTE** (Synthetic Minority Over-sampling Technique) and **ADASYN** (Adaptive Synthetic Sampling), could be evaluated for their effect on model performance.

4. **Model Interpretability:** Integrating explainability frameworks such as **SHAP (SHapley Additive exPlanations)** and **LIME (Local Interpretable Model-agnostic Explanations)** would allow the insurance company to understand the specific factors driving each individual prediction, increasing trust and regulatory compliance.

REFERENCES

1. Baesens, B., Van Gestel, T., Viaene, S., Stepanova, M., Suykens, J., and Vanthienen, J. (2009). 'Benchmarking State-of-the-Art Classification Algorithms for Credit Scoring', *Journal of the Operational Research Society*, Vol. 54, No. 6, pp. 627–635.

2. Charpentier, A., Denuit, M., and Elie, R. (2018). 'Machine Learning in Insur- ance', *SSRN Electronic Journal*. Available at: `https://ssrn.com/abstract=3242672`.

3. Chawla, N.V., Bowyer, K.W., Hall, L.O., and Kegelmeyer, W.P. (2002). 'SMOTE: Synthetic Minority Over-sampling Technique', *Journal of Artificial Intelligence Research*, Vol. 16, pp. 321–357.

4. Chen, T. and Guestrin, C. (2016). 'XGBoost: A Scalable Tree Boosting System', *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, San Francisco, CA, USA, pp. 785–794.

5. Géron, A. (2019). *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow*. 2nd edn. O'Reilly Media, Sebastopol, CA.

6. Geurts, P., Ernst, D., and Wehenkel, L. (2006). 'Extremely Randomized Trees', *Machine Learning*, Vol. 63, No. 1, pp. 3–42.

7. Guelman, L., Guillen, M., and Perez-Marin, A.M. (2014). 'Uplift Random Forests', *Cybernetics and Systems*, Vol. 45, No. 3, pp. 230–248.

8. Hartfeil, G. (1996). 'Bank One Implements Optimal Database-Marketing Programs', *Interfaces*, Vol. 26, No. 6, pp. 1–13.